Andrei Stefan Bejgu (he/him)

Date of Birth: 16-11-1995
Location: Rome, Italy
Languages: English (Fluent), Italian (Native), Romanian (Native)

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Senior AI Applied Scientist at Syllotips, where I work on the engine behind its agents: how they plan, which tools they call and how they cite their sources, and how to find where they fail and keep improving them. Before that I spent 8 years at Babelscape, growing from software engineer to NLP researcher. Along the way I earned an industrial Ph.D. in AI at Sapienza under Roberto Navigli, published at ACL, EMNLP, EACL and NAACL, and co-first-authored LLM-Oasis, on checking whether what an LLM writes is true.

Experience & Selected Projects

Syllotips — *Remote*

Senior AI Applied Scientist — Oct. 2025–Present

I build the agents of Syllotips: LangGraph agents that answer from a company's knowledge, act on its systems through MCP, and improve as experts correct them.

- Built the **deep-research agent** of the Syllotips agent stack and the **harness** it runs in: per-turn tool budgets, bounded replanning, a Redis-backed scratch filesystem that keeps large MCP results as file handles instead of truncating them, and source memory that carries across turns.
- Built the sandboxed **code interpreter** for spreadsheet analysis and report generation. Runs that work become reusable recipes; with missing data it stops with a blocker instead of inventing numbers.
- Built the **guardian agent** that turns a failed run into a fix: it inspects the failing agent's tools read-only, attributes the failure, and proposes the smallest change to its knowledge or procedure for an expert to approve.
- Cut LLM cost and latency with **prompt caching** across turns and tenants, and by ending tool loops once the plan is settled.
- Set up Arize Phoenix tracing, a benchmark comparing ReAct and plan-and-execute agents on realistic enterprise knowledge bases, and an 80% coverage gate on new code in CI.

Babelscape — *Rome, Italy*

NLP Researcher — Oct. 2021–Sep. 2025

I led advanced NLP research translating state-of-the-art insights into production AI products, with a focus on retrieval-augmented systems and multilingual LLM fine-tuning.

- Designed and developed real-time retrieval pipelines, maximizing precision and efficiency for Vera, a multilingual fact-checking system.
- **Distributed** (Slurm) fine-tuning of open-source LLMs for domain-specific tasks such as **text2sql and claim-extraction/verification**.
- Contributed to the supervised-fine-tuning stage of **Minerva**, the first Italian LLM.

NLP Engineer — Jun. 2020–Oct. 2021

- Designed and deployed Semantic Paths, a **Multilingual Semantic Search** engine for scientific articles written in Kotlin.
- Developed advanced **Word Sense Disambiguation** models for large-scale multilingual processing.

Software Engineer — Jul. 2017–Jun. 2020

- Developed myKnowledgeGraph, a **Knowledge Graph Extraction** and visualization system for enterprise data.
- Designed the company's **Java NLP Pipeline**, maximizing speed, accuracy, and scalability of multilingual text processing.

CINI - Consorzio Interuniversitario Nazionale per l'Informatica — *Rome, Italy*

Software Developer — Dec. 2016–Jan. 2017

Created an advanced tool for lexicon-semantic stylometric analysis, enabling efficient processing of large-scale texts.

Education

Ph.D. in Artificial Intelligence
Sapienza University of Rome Jan. 2025
Focus: Explored multilingual NLP by applying lexical semantics to modern systems. Improved domain-specific tasks, looked into complex linguistic patterns, and worked on word- and sentence-level representations in RAG systems and LLMs. This was a joint industrial Ph.D., combining academic research with industry-driven projects. Created context-aware, domain-specific embeddings that boosted retrieval performance. –**Advisor:** Prof. R. Navigli

MSc in Computer Science
*Sapienza University of Rome –**Grade**: 110/110 with honors* Nov. 2021
Thesis: Exploring language-agnostic latent spaces for sentences. –**Advisor:** Prof. R. Navigli

BSc in Computer Science
*Sapienza University of Rome –**Grade**: 107/110* Jun. 2019
Thesis: Knowledge-Based Word Sense Disambiguation leveraging synset embeddings. –**Advisor:** Prof. R. Navigli

Publications

[1] A. Scirè*, **A. S. Bejgu***, S. Tedeschi, K. Ghonim, F. Martelli, and R. Navigli. "Truth or Mirage? Towards End-to-End Factuality Evaluation with LLM-OASIS". In: *Computational Linguistics* (2025).

[2] L. Moroni, T. Bonomo, L. Gioffré, L. Xu, D. Fedele, L. Colosi, **A. S. Bejgu**, A. Scirè, and R. Navigli. "What we Learned from Continually Training Minerva: a Case Study on Italian". In: *CLiC-it*. 2025.

[3] L. Moroni, G. Puccetti, P.-L. H. Cabot, **A. S. Bejgu**, A. Miaschi, E. Barba, F. dell'Orletta, A. Esuli, and R. Navigli. "Optimizing LLMs for Italian: Reducing Token Fertility and Enhancing Efficiency Through Vocabulary Adaptation". In: *Findings of NAACL*. 2025.

[4] K. Ghonim, **A. S. Bejgu**, and R. Navigli. "Conceptpedia: Leveraging Traditional Approaches for Multimodal Image-Text Representation". In: *EMNLP*. 2025.

[5] **A. S. Bejgu**, E. Barba, L. Procopio, A. Fernández-Castro, and R. Navigli. "Word Sense Linking: Disambiguating Outside the Sandbox". In: *ACL*. 2024.

[6] F. Molfese, **A. S. Bejgu**, S. Tedeschi, S. Conia, and R. Navigli. "CroCoAlign: A Cross-Lingual, Context-Aware and Fully-Neural Sentence Alignment System for Long Texts". In: *EACL*. 2024.

[7] F. Martelli, **A. S. Bejgu**, et al. "XL-WA: a Gold Evaluation Benchmark for Word Alignment in 14 Language Pairs". In: *CLiC-it*. 2023.

Other Activities

Reviewer Activity: Reviewer for EACL (2023), ACL (2023, external reviewer), EMNLP (2023, external reviewer), ICLR (2024, external reviewer), and ACL ARR (2024 and 2025).
Teaching Activity at Sapienza University of Rome:

- **Natural Language Processing Course (2022–2023)**: Designed and led practical lab sessions and assignments, guiding master's students through state-of-the-art NLP techniques.
- **Hot Topics in NLP (2023–2024)**: Developed and managed one of the three major course assignments, providing hands-on exposure to cutting-edge advancements in NLP.

Scholarships and Awards

- Best Employee Award, Babelscape 2021
- Merit Scholarship Winner, LazioDiSCo.it 2018–2020